Web Introduction

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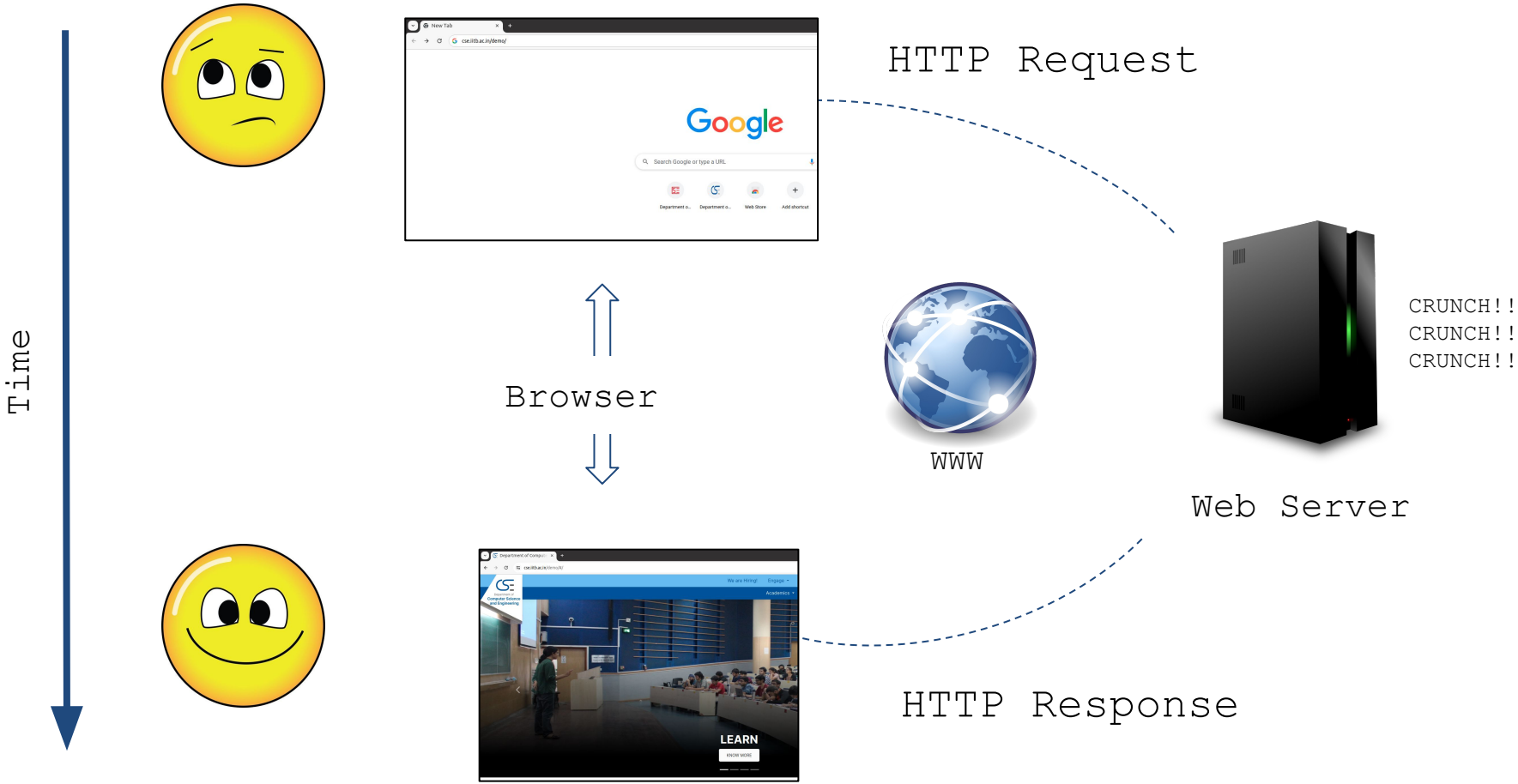
World Wide Web (WWW)

- A service provided over Internet, enabling users to access and interact with a vast array of information and services

Terminology

- Web Server: a software application hosted on a machine, that delivers web content/service
  - Browser requests arrive here
  - Serves **static** content
  - Examples: Apache, Nginx, Microsoft IIS
- Web Client (Browser): Application running on user's device
  - Sends out requests to server, processes received responses and renders web pages

- Web Application: a software program or set of programs that run on a machine
  - Help **dynamically** generates content in response to requests
  - Involve server-side scripting, database interactions, and complex business logic
  - E.g. Amazon, Facebook, Google docs, BodhiTree etc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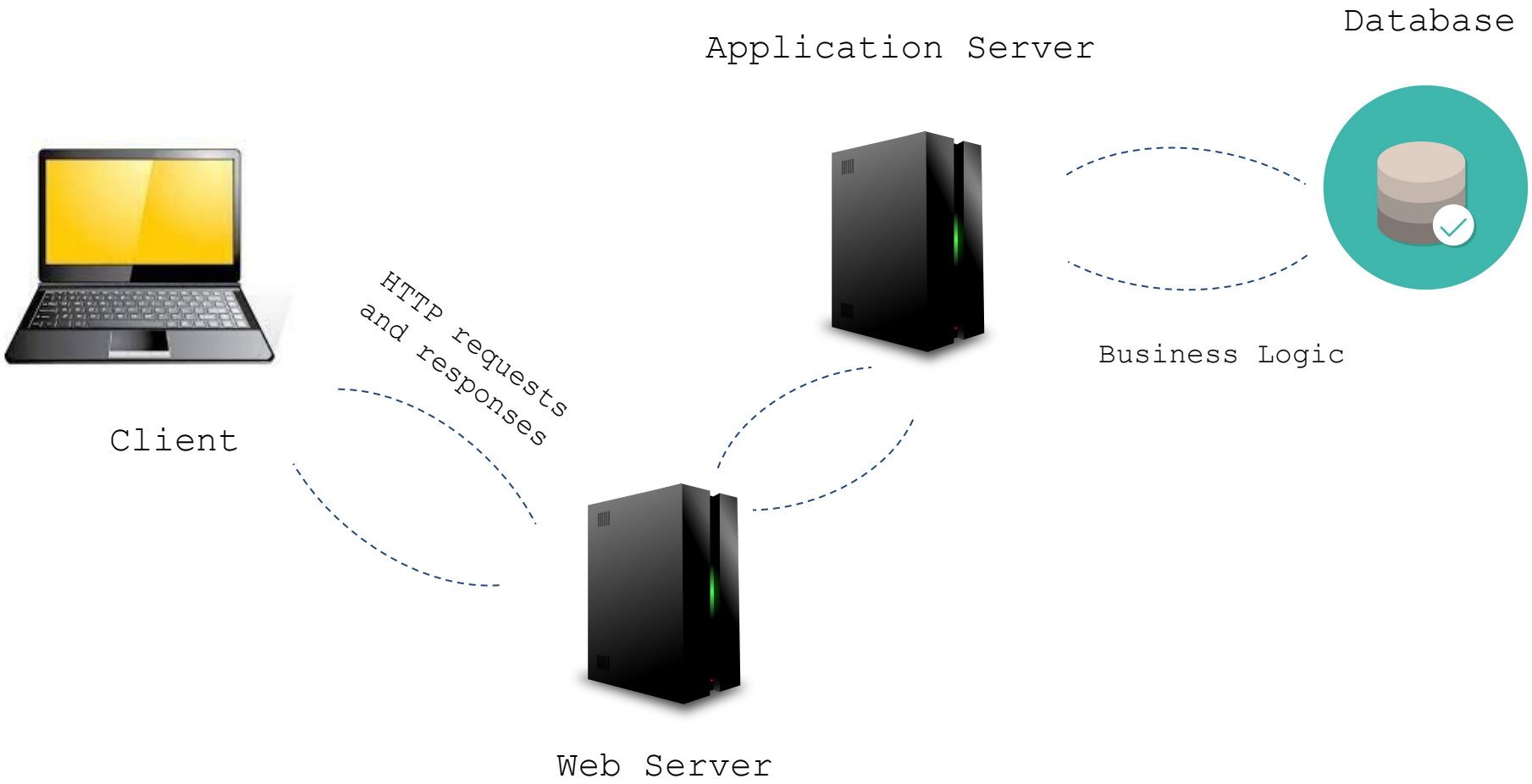
Database
Application Server
HTTP requests
and responses
Business Logic
Client
Web Server

Web Client-Server Interaction

Typical Workflow (At high level)

- Web Request:
  - A user enters a web address (URL) in browser
  - One or more requests sent to web server hosting website
- Processing:
  - Server/Application processes each request
    - May run server-side scripts, query databases, or performing other tasks to generate a response

- Response:
  - The generated response(s) sent back to user's browser
- Delivery:
  - Browser interprets response(s) and displays web page
- Repeat above steps:
  - User interacts with the page causing more requests to be sent out
  - Receives corresponding responses and sees new information, and so on

Outline

- What constitutes a webpage?
- What goes on inside a Browser (web client)?
  - What standard security mechanisms implemented?
- How do client and server communicate?
  - HTTP/HTTPs protocol
  - Session Management via cookies and tokens
- How does a web server process requests and generate responses?
  - Static vs Dynamic content